

Dr. Babasaheb Ambedkar Technological University

Supplementary Summer Examination – 2025

Course: B. Tech.

Branch: Common to all branches

Semester: I/II

Subject Code & Name: **BTES103 Engineering Mechanics**

Max Marks: 60

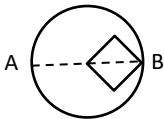
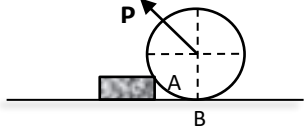
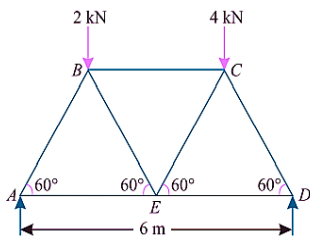
Date: 23-07-2025

Duration: 3 Hrs.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question **No. 1 will be compulsory and include objective-type questions.**
3. Candidates are required to **attempt any four questions from Q. No. 2 to Q. No. 6.**
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

Q. 1	Objective type questions. (Compulsory Question)	(Level /CO)	Marks
			12
1	The primary purpose of a support in a structure is to: a) Provide resistance to applied loads b) Increase load intensity c) Add weight to the structure d) Allow free movement	CO1	01
2	Which type of support allows rotation but prevents translation in any direction a) Fixed support b) Roller support c) Hinged support d) Free support	CO1	01
3	A roller support primarily resists forces in which direction? a) Horizontal b) Vertical c) Both a & b d) Rotational	CO2	01
4	If a system of forces acting on a body produces no motion, the system is said to be in: a) Motion b) Dynamic state c) Equilibrium d) Inertia	CO2	01
5	When two or more forces acting at a point are combined into a single force, it is called: a) Equilibrant b) Resultant force c) Couple c) Load factor	CO3	01
6	The unit of moment in the SI system is: a) Newton b) N-m c) Pa d) Joule	CO3	01
7	According to Lami's theorem, the ratio of each force to the sine of the angle opposite to it is: a) Equal b) Different c) Zero d) Negative	CO4	01
8	Varignon's theorem is also known as: a) Law of moments b) Principle of superposition c) Principle of transmissibility d) Law of inertia	CO4	01
9	The unit of moment of inertia in SI units is: a) m ⁴ b) kg m ⁴ c) m ² d) N-m	CO4	01
10	Which section has the highest moment of inertia for the same area? a) Circular section b) Square section c) Rectangular section d) I-section	CO5	01
11	In D'Alembert's principle, the system is considered to be in: A) Dynamic equilibrium B) Rotational equilibrium C) Static equilibrium D) Translational equilibrium	CO5	01
12	The SI unit of power is: a) Joule b) Watt c) Newton d) Horsepower	CO1	01

Q. 2	Solve the following.		12
A)	Classify the system of forces with neat sketches & explain them in detail?	CO1	06
B)	What are the types loads and explain them in detail with neat sketches?	CO1	06
Q.3	Solve the following.		12
A)	What are the types of 2D supports and explain them in detail with neat sketches?	CO2	06
B)	A square hole is punched out of circular lamina, the diagonal of the square being the radius of the circle as shown in Fig 2.1 (A) below Find the center of gravity of the remainder, if 'r = 4 cm' as the radius of the circle. Fig 2.1 (A) See this figure at the last page.	CO2	06
Q. 4	Solve Any Two of the following.		12
A)	A Uniform wheel of 1200 mm diameter weighing 10 kN rests against a rigid rectangular block of 300 mm height as shown in figure below. Find the least force (P) through the centre of the wheel required just to turn the wheel over the corner A of the block. Fig 2.1 (B) See this figure at the last page.	CO3	06
B)	State and explain the Lami's theorem and prove that?	CO3	06
C)	Define: a) Static Friction, b) Dynamic Friction, c) Angle of Friction, e) Angle of repose. What are the Coulomb's laws of dry friction?	CO3	06
Q.5	Solve Any Two of the following.	CO4	12
A)	A ladder 5 meters long rests on a horizontal ground and leans against a smooth vertical wall at an angle 70° with the horizontal. The weight of the ladder is 900 N and acts at its middle. The ladder is at the point of sliding, when a man weighing 750 N stands on a rung 1.5 meter from the bottom of the ladder. Calculate coefficient of friction between ladder & the floor.	CO4	06
B)	A Warren girder consisting of seven members each of 3 m length freely supported at its end points. The girder is loaded at B and C as shown. Find the forces in all the members of the girder, indicating whether the force is compressive or tensile. Fig 3.1 (C) See this figure at the last page.	CO4	06
C)	The equation of motion of an engine is given by $s = 2t^3 - 6t^2 - 5$, where (s) is in meters and (t) in seconds. Find (i) displacement & acceleration when velocity is zero; & (ii) displacement & velocity when acceleration is zero.	CO4	06
Q. 6	Solve Any Two of the following.		12
A)	A Passenger train 300 m long, moving with a velocity of 108 kmph, overtakes a goods train moving on a parallel path in the same direction, completely in 45 seconds. If the length of the goods train is 250 m, Determine the speed of the goods train?	CO5	06
B)	A ball of mass 10 kg moving with a velocity of 20 m/s impinges directly on a ball of mass 20 kg at rest. The first ball, after impinging, comes to rest. Find the velocity of the second ball after the impact and the coefficient of restitution.	CO5	06
C)	A motorist travelling at a speed of 70 kmph suddenly applies a breaks and halts after skidding 50 m. Determine A) The time required to stop the car	CO5	06
 Fig 2.1 (A)  Fig 2.1 (B)  Fig 3.1 (C) *** End ***			